

Documenting a network experiment

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Efficiency and resource measurement

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Uncertainty and failure analysis

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the selected result for Documenting a network experiment. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Documenting a network experiment. Proposed explanation: “A smaller MSE is an accuracy percentage”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a new record independently

A separate run R17 records baseline $a = 1$ and checkpoints $a = 1.5$ and $a = 1.75$. It uses prediction $\max(0, a \cdot x)$ and one validation case V9 with $x = 4$, $y = 8$. Compute baseline and checkpoint predictions and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Write the replay settings

For NN32-A, record the prediction rule, initial weight, bias policy, learning rate, update count, loss, training/validation IDs, data order or shuffling policy, and checkpoint selection rule. Name one environment field still to record when you run it.

My reasoning and evidence

6. W2 Trace the selected result

Calculate both weight updates and give each training and validation MSE. Name the selected checkpoint

and baseline validation MSE. How many independent validation examples support the comparison?

My reasoning and evidence

7. W3 Audit the artifact record

A folder contains only best-model.json and a screenshot of 0.5625. Identify four missing artifacts or records and explain what a content hash can and cannot verify.

My reasoning and evidence

8. W4 Audit a new record independently

A separate run R17 records baseline $a = 1$ and checkpoints $a = 1.5$ and $a = 1.75$. It uses prediction $\max(0, a \cdot x)$ and one validation case V9 with $x = 4$, $y = 8$. Compute baseline and checkpoint predictions and squared errors, choose by lowest validation MSE, and write a bounded conclusion. Final evaluation and runtime have not been measured.

My reasoning and evidence

9. W5 Investigate a replay mismatch

A partner gets different results after changing preprocessing and replacing the validation file. They keep the old run ID and overwrite the old score. Explain what should be preserved and how the revised work should be recorded.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled documenting a network experiment evidence sheet to Image-pattern Network Investigation. Show how you can record a network experiment with enough detail to replay its calculations. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Documenting a network experiment

Document checkpoint selection so another team can reconstruct every reported number.

Run R8 predicts $\text{ReLU}(a \times x)$. Validation V4 has $x = 2$, target 3. Baseline $a = 0.5$; checkpoints $a = 1$ and 1.25; choose minimum validation squared error.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Validation squared loss by checkpoint

squared error

baseline $a = 0.5$  4

checkpoint $a = 1$  1

checkpoint $a = 1.25$  0.25

Bar length shows the supplied value; the marked baseline is zero.

Documenting a network experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Baseline prediction

At $a = 0.5$, prediction 1, error2 = 4.



2. Checkpoint comparison

At $a = 1$, prediction 2, loss 1; at 1.25, prediction 2.5, loss 0.25.



3. Select and record

Choose $a = 1.25$; save R8, inputs, formula, checkpoint values, loss rule and validation rule.

CHANGE ONE THING

A teammate calls V4 a final test after using it to select $a=1.25$.

Explain the changed result using the matching lesson.

What additional evidence is needed for a final performance claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Document checkpoint selection so another team can reconstruct every reported number.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Choose $a = 1.25$; save R8, inputs, formula, checkpoint values, loss rule and validation role.

1. Checkpoint comparison
2. Select and record
3. Baseline prediction

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Select and record Choose $a = 1.25$; save R8, inputs, formula, checkpoint values, loss rule and validation role. A reproducible record includes the actual input, formula, parameters and scoring rule, together with each dataset's role. The validation case selects the best supplied checkpoint, so it is part of development. Save that selection history and freeze the checkpoint before evaluating fresh final cases. Renaming the same case a test does not change how it was used.

Practice 2: explain the deciding evidence

Changed-case practice: A teammate calls V4 a final test after using it to select $a=1.25$. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.

3. Correct its role to validation; it was used for selection.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Correct its role to validation; it was used for selection. Changing the label on a dataset does not make its evidence untouched.

Repair a missing building block

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10ms/case not necessarily a single-request latency? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
